

# Solutions to Axler's *Linear Algebra Done Right*, 4<sup>th</sup> Edition

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## 7D Isometries, Unitary Operators, and Matrix Factorization

### Exercise 7D1

Suppose  $\dim V \geq 2$  and  $S \in \mathcal{L}(V, W)$ . Prove that  $S$  is an isometry if and only if  $Se_1, Se_2$  is an orthonormal list in  $W$  for every orthonormal list  $e_1, e_2$  of length two in  $V$ .

### Exercise 7D2

Suppose  $T \in \mathcal{L}(V, W)$ . Prove that  $T$  is a scalar multiple of an isometry if and only if  $T$  preserves orthogonality.

**Axler Note.** The phrase “ $T$  preserves orthogonality” means that  $\langle Tu, Tv \rangle = 0$  for all  $u, v \in V$  such that  $\langle u, v \rangle = 0$ .

### Exercise 7D3

- (a) Show that the product of two unitary operators on  $V$  is a unitary operator.
- (b) Show that the inverse of a unitary operator on  $V$  is a unitary operator.

**Axler Note.** This exercise shows that the set of unitary operators on  $V$  is a group, where the group operation is the usual product of two operators.

### Exercise 7D4

Suppose  $\mathbb{F} = \mathbb{C}$  and  $A, B \in \mathcal{L}(V)$  are self-adjoint. Show that  $A + iB$  is unitary if and only if  $AB = BA$  and  $A^2 + B^2 = I$ .

### Exercise 7D5

Suppose  $S \in \mathcal{L}(V)$ . Prove that the following are equivalent.

- (a)  $S$  is a self-adjoint unitary operator.
- (b)  $S = 2P - I$  for some orthogonal projection  $P$  on  $V$ .
- (c) There exists a subspace  $U$  of  $V$  such that  $Su = u$  for every  $u \in U$  and  $Sw = -w$  for every  $w \in U^\perp$ .

### Exercise 7D6

Suppose  $T_1, T_2$  are both normal operators on  $\mathbb{F}^3$  with 2, 5, 7 as eigenvalues. Prove that there exists a unitary operator  $S \in \mathcal{L}(\mathbb{F}^3)$  such that  $T_1 = S^*T_2S$ .

**Exercise 7D7**

Give an example of two self-adjoint operators  $T_1, T_2 \in \mathcal{L}(\mathbb{F}^4)$  such that the eigenvalues of both operators are 2, 5, 7 but there does not exist a unitary operator  $S \in \mathcal{L}(\mathbb{F}^4)$  such that  $T_1 = S^*T_2S$ . Be sure to explain why there is no unitary operator with the required property.

**Exercise 7D8**

Prove or give a counterexample: If  $S \in \mathcal{L}(V)$  and there exists an orthonormal basis  $e_1, \dots, e_n$  of  $V$  such that  $\|Se_k\| = 1$  for each  $e_k$ , then  $S$  is a unitary operator.

**Exercise 7D9**

Suppose  $\mathbb{F} = \mathbb{C}$  and  $T \in \mathcal{L}(V)$ . Suppose every eigenvalue of  $T$  has absolute value 1 and  $\|Tv\| \leq \|v\|$  for every  $v \in V$ . Prove that  $T$  is a unitary operator.

**Exercise 7D10**

Suppose  $\mathbb{F} = \mathbb{C}$  and  $T \in \mathcal{L}(V)$  is a self-adjoint operator such that  $\|Tv\| \leq \|v\|$  for all  $v \in V$ .

- (a) Show that  $I - T^2$  is a positive operator.
- (b) Show that  $T + i\sqrt{I - T^2}$  is a unitary operator.

**Exercise 7D11**

Suppose  $S \in \mathcal{L}(V)$ . Prove that  $S$  is a unitary operator if and only if

$$\{Sv \mid v \in V \text{ and } \|v\| \leq 1\} = \{v \in V \mid \|v\| \leq 1\}.$$

**Exercise 7D12**

Prove or give a counterexample: If  $S \in \mathcal{L}(V)$  is invertible and  $\|S^{-1}v\| = \|Sv\|$  for every  $v \in V$ , then  $S$  is unitary.

**Exercise 7D13**

Explain why the columns of a square matrix of complex numbers form an orthonormal list in  $\mathbb{C}^n$  if and only if the rows of the matrix form an orthonormal list in  $\mathbb{C}^n$ .

### Exercise 7D14

Suppose  $v \in V$  with  $\|v\| = 1$  and  $b \in \mathbb{F}$ . Also suppose  $\dim V \geq 2$ . Prove that there exists a unitary operator  $S \in \mathcal{L}(V)$  such that  $\langle Sv, v \rangle = b$  if and only if  $|b| \leq 1$ .

### Exercise 7D15

Suppose  $T$  is a unitary operator on  $V$  such that  $T - I$  is invertible.

(a) Prove that  $(T + I)(T - I)^{-1}$  is a skew operator (meaning that it equals the negative of its adjoint).

(b) Prove that if  $\mathbb{F} = \mathbb{C}$ , then  $i(T + I)(T - I)^{-1}$  is a self-adjoint operator.

**Axler Note.** The function  $z \mapsto i(z + 1)(z - 1)^{-1}$  maps the unit circle in  $\mathbb{C}$  (except for the point 1) to  $\mathbb{R}$ . Thus (b) illustrates the analogy between the unitary operators and the unit circle in  $\mathbb{C}$ , along with the analogy between the self-adjoint operators and  $\mathbb{R}$ .

### Exercise 7D16

Suppose  $\mathbb{F} = \mathbb{C}$  and  $T \in \mathcal{L}(V)$  is self-adjoint. Prove that  $(T + iI)(T - iI)^{-1}$  is a unitary operator and 1 is not an eigenvalue of this operator.

### Exercise 7D17

Explain why the characterizations of unitary matrices given by 7.57 hold.

### Exercise 7D18

A square matrix  $A$  is called **symmetric** if it equals its transpose. Prove that if  $A$  is a symmetric matrix with real entries, then there exists a unitary matrix  $Q$  with real entries such that  $Q^*AQ$  is a diagonal matrix.

**Exercise 7D19**

Suppose  $n$  is a positive integer. For this exercise, we adopt the notation that a typical element  $z$  of  $\mathbb{C}^n$  is denoted by  $z = (z_0, z_1, \dots, z_{n-1})$ . Define linear functionals  $\omega_0, \omega_1, \dots, \omega_{n-1}$  on  $\mathbb{C}^n$  by

$$\omega_j(z_0, z_1, \dots, z_{n-1}) = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} z_m e^{-2\pi i j m / n}.$$

The **discrete Fourier transform** is the operator  $\mathcal{F} : \mathbb{C}^n \rightarrow \mathbb{C}^n$  defined by

$$\mathcal{F}z = (\omega_0(z), \omega_1(z), \dots, \omega_{n-1}(z)).$$

- (a) Show that  $\mathcal{F}$  is a unitary operator on  $\mathbb{C}^n$ .
- (b) Show that if  $(z_0, \dots, z_{n-1}) \in \mathbb{C}^n$  and  $z_n$  is defined to equal  $z_0$ , then

$$\mathcal{F}^{-1}(z_0, z_1, \dots, z_{n-1}) = \mathcal{F}(z_n, z_{n-1}, \dots, z_1).$$

- (c) Show that  $\mathcal{F}^4 = I$ .

**Axler Note.** The discrete Fourier transform has many important applications in data analysis. The usual Fourier transform involves expressions of the form  $\int_{-\infty}^{\infty} f(x) e^{-2\pi i t x} dx$  for complex-valued integrable functions  $f$  defined on  $\mathbb{R}$ .

**Exercise 7D20**

Suppose  $A$  is a square matrix with linearly independent columns. Prove that there exist unique matrices  $R$  and  $Q$  such that  $R$  is lower triangular with only positive numbers on its diagonal,  $Q$  is unitary, and  $A = RQ$ .